

NANA APPIAGYEI POKU

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EDUCATION

McMaster University

Hamilton, ON

Bachelor of Applied Science in Computer Science, Minor in Finance

Expected April 2028

- **GPA:** 3.5/4.0 | **Awards:** ACCPI Award, John C. Holland Award (Black Community Leadership and Academic Excellence)

EXPERIENCE

Software Engineering Intern

Summer 2026

Microsoft

Redmond, WA

- Built and shipped full-stack features for a product serving **500M+** users using **TypeScript, React, TanStack Query**, and **Redux**.
- Improved reliability and scalability of production services by writing, reviewing, and debugging high-quality code with **Git**, **CI/CD** pipelines, and **Agile** workflows.
- Exceeded project expectations by designing a more efficient data-loading strategy for a feature, improving query speed by **50%** over the original plan while fetching **10x** more data.

Machine Learning Research Assistant (Co-op)

Apr 2025 – Sep 2025

McMaster Software Labs (ECE Dept.)

Hamilton, ON

- Reduced neural network weight matrix storage by **20%** with no loss in model accuracy by designing custom matrix compression algorithms.
- Built **Python** data-analysis pipelines (**Pandas, Matplotlib, Seaborn**) that quantified entropy, variance, and distributions to guide compression decisions.
- Identified statistical patterns in neural-network weights that improved storage and compression efficiency.

Website Developer

Aug 2025 – Present

Blackprint

Hamilton, ON

- Build responsive, **WCAG 2.1** accessible web pages using **TypeScript, JavaScript, HTML, CSS**, and **Bootstrap** that deliver a consistent experience across devices.
- Improve usability and engagement by developing interactive components (navigation menus, forms, modals) in **TypeScript**.

NSBE IDEA Conference Co-Coordinator

Sep 2024 – Present

National Society of Black Engineers

Hamilton, ON

- Coordinate logistics, mentorship, and alumni networking for an equity, diversity, and inclusion engineering summit serving **100+** attendees.

PROJECTS

Accessibility-Based Web App | Python, JavaScript, Whisper API

DeltaHacks XI

- Implemented real-time speech translation with the **Whisper API**, achieving **95%+** accuracy across multiple languages.
- Built a tone-classification feature for nuanced sentiment detection with low-latency audio-to-text processing.

Stock Predictor | Python, TensorFlow, PyTorch, Tkinter

- Developed a stock-prediction model using **Yahoo Finance** and **Alpha Vantage** APIs with engineered features (RSI, MACD, moving averages).
- Improved forecast accuracy by **12%** over baseline models and visualized trends with **Matplotlib, Plotly**, and **Seaborn**.

Augmented Reality Mobile App | React, Python, OpenCV, MediaPipe

- Built a real-time AR try-on experience using **MediaPipe** pose estimation and body-landmark tracking to anchor garments on users.
- Integrated **OpenCV** camera processing with a **React** frontend featuring catalog view, garment selection, and live preview.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Go, JavaScript, TypeScript, SQL, Ruby, HTML/CSS

Frameworks & Libraries: React, Redux, TanStack Query, Django, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Bootstrap, REST APIs

Databases: MySQL, PostgreSQL, MongoDB

Developer Tools: Git, Linux, Node.js, VS Code, IntelliJ, MATLAB

RELEVANT COURSEWORK

Data Structures, Algorithms, Software Design, Object-Oriented Programming (Java), Web Development, Internet Protocols (TCP/IP), Machine Learning, Linear Algebra, Probability & Statistics, Differential Equations